

Properties of Crude Oil Responsible for Pipeline Accidents

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Problem Statement

What qualities of crude oil lead to pipeline accidents and what are the related regulations?

Part A

What physical or chemical properties of crude oil could be responsible for pipeline accidents?

Part B – Regulations & Implications

- What regulations on those properties would you recommend?
- Have we gotten better at regulating those properties over time?
- What implications on the oil market would these regulations have?
- How can success of these regulations be quantified in the short term and long term?

The background of the slide is an abstract, marbled pattern in shades of deep blue and black. The pattern consists of swirling, organic shapes that resemble liquid or smoke, creating a complex and textured visual field.

Part A – Physical or Chemical Properties of Crude Oil that could be Responsible for Pipeline Accidents

Crude Oil

To determine what underlying properties of crude oil could be contributing to pipeline accidents, we need to first understand the composition of crude oil. This includes its chemical composition, its physical properties, and any additional qualities that could be related to pipeline incidents.

General

Crude oil is an unrefined petroleum-based product that is primarily composed of hydrocarbon gases and liquids. A sample breakdown of Bakken crude oil, considered a light crude oil, is given in Appendix A [1]. The classifications of crude oil’s physical and chemical properties can be seen below. Crude oil is refined into a variety of products, such as gasoline, kerosene, asphalt, or plastics.

Physical Properties

The weight of crude oils is measured on an arbitrary scale based on their relative gravity/density to water, where water has 10 API gravity. Heavy crudes have an API of 22 or below, lighter crudes generally exceed 38 API. [3]

CLASS A: LIGHT, VOLATILE OILS [2]

- Highly fluid
- Often clear
- Spread rapidly on solid or water surfaces
- Have a strong odor
- High evaporation rate
- Usually flammable

CLASS C: HEAVY, STICKY OILS

- Viscous
- Sticky or tarry
- Brown or black

CLASS B: NON-STICKY OILS

- Waxy or oily
- Less toxic
- Penetrate porous substances
- Evaporation of volatiles leads to Class C or D residue

CLASS D: NON-FLUID OILS

- Relatively non-toxic
- Do not penetrate porous substances
- Black or dark brown

Chemical Properties

Crude Oil is also known for containing varying concentrations of sulfur.

SWEET: <0.5% sulfur content

SOUR: >=0.5% sulfur content

Pipeline Accidents

The PHMSA lists 10 unique causes for pipeline failures along with notable characteristics and what percentage they make up of total pipeline incidents [5]. Given these failure types, we want to determine which could be potentially relevant to our analysis so that we can filter our data accordingly.

Pipeline Failure Cause	Description	1998-2017 Percentage Pipeline Accidents	Weight-relevant?	Sulfur-relevant?
External Corrosion	Due to environmental conditions on the outside of the pipe	8 %	No	No
Internal Corrosion	Contaminants naturally contained within the transported commodity, such as hydrogen sulfide, carbon dioxide, or other chemicals	12 %	No	Yes
Stress Corrosion Cracking	Environmentally-assisted cracking that reduces the pressure-carrying capability of the pipe	1 %	Maybe	Maybe
Selective Seam Corrosion	Localized corrosion along specific welding points. Generally, has been fixed since 1970 due to upgrade in materials.	1 %	No	No
Excavation Damage	Excavator damaging exterior of pipeline	15 %	No	No
Natural Force Damage	Naturally occurring events, such as earth movement or heavy rains.	9 %	No	No
Other Outside Force Damage	Vehicle or equipment contact; vandalism; damage caused by accidents, etc.	<1	No	No
Material/Weld Failure	Impurities remaining in the molten steel upon manufacturing.	Not Provided	Maybe	Maybe
Equipment Failure	Pipeline component other than the pipe. E.g., pumps and compressors, meters and metering equipment, valves	Not Provided	Yes	Maybe
Incorrect Operation	Direct or indirect cause of a pipeline failure; e.g., wrong valve open, overfilling a tank, over-pressuring a piece of equipment	Not Provided	No	No

Key Takeaways

PHYSICAL PROPERTIES

Focus on accidents in which the weight of the substance could have exacerbated stress corrosion, or material/equipment failure.

CHEMICAL PROPERTIES

Focus on incidents in which internal corrosion occurred, or material failures where the substance could have deteriorated the performance of the equipment/system.

Dataset Collection

We need to source data that contains information on pipeline incidents. The ideal dataset will contain detailed information on the substances that are being transported and the cause of the accident. Additional information about pipeline specifications could be valuable.

Canadian Datasets*

The Canada Energy Regulator (CER) (formally the National Energy Board) regulates pipelines that cross inter-provincial or international boundaries; pipelines that are intra-provincial are regulated by each individual province [6]. The Onshore Pipeline Regulations (OPR) were created by the CER to ensure companies are meeting pipeline safety requirements.

American Datasets

The Pipeline and Hazardous Materials Safety Administration (PHMSA) is responsible for regulating and ensuring the safe and secure movement of hazardous materials within the United States [7].

Label	Dataset Name	Description	Date Range	# Records	Crude Oil Types	Weight-specific info?	Sulfur-specific info?	Accident-specific info
A	Pipeline-incidents-comprehensive-data.csv	Pipeline incidents as defined by OPR and Processing Plant regulations	2008 to Present	1856	Crude Oil, Crude Oil (Heavy, Light, Sour, and Sweet)	No	No – grouped together	What happened: damage, deterioration, construction, etc.; Incident type: fire, release of substance, etc.
B	PODSdb_MDOTW_VW_OCCURRENCE_PUBLIC.csv	Incidents at CER-regulated pipelines and facilities as defined by OPR	2008 to current	3265	Crude Oil, Crude Oil-Sour	No	Sour vs. not sour	Classification of Occurrence (encoded); occurrence summary
C	Accident_hazardous_liquid_pre1986.txt	Incidents within Hazardous Liquid and Carbon Dioxide Pipelines from 1968 to 1985	1968 to 1985	4734	Crude Oil	No	No	PRIMARY_CAUSE: corrosion, equipment, defective pipe, etc.
D	Accident_hazardous_liquid_1986_jan2002.txt	^ [1986:2002]	1986 to 2002	3095	Crude Oil	No	No	CAUS: corrosion, failed pipe, malfunction, etc. ; CAUSO: breakdown of CAUS; NARRATIVE
E	Accident_hazardous_liquid_jan2002_dec2009	^ [2002:2009]	2002 to 2009	3031	Crude Oil, Crude Oil Heavy, Crude Oil Sour, Crude Oil / Location, and more	Not explicitly; can be roughly extrapolated	Not explicitly; can be roughly extrapolated	Small spills: GEN_CAUSE (equipment, corrosion, etc.) Big spills: CAUSE (more detailed than small spills); NARRATIVE
F	Accident_hazardous_liquid_jan2010_present	^ [2010:2023]	2010 to 2023	5248	Crude Oil	No	No	CAUSE: corrosion, equipment, manual error, etc. CAUSE_DETAILS: more detailed than above; NARRATIVE
G	Accident_gravity_reporting_regulated_jul2020_present.txt	Transportation incidents of hazardous liquids flowing through gravity pipelines.	2021 to 2023	297	Crude Oil (Nothing else)	No	No	CAUSE: corrosion, equipment, manual error, etc. CAUSE_DETAILS: more detailed than above; NARRATIVE

*Omitting pipeline transportation occurrence data from Jan 1979 to 1990 as the TSB does not warrant the quality, accuracy, reliability or completeness of its data [8]

Approach

Before diving into a dataset, we need to ensure that we're taking a logical approach to the problem that encompasses the possible scenarios / outcomes.

Approach #1 – Crude Oil vs. NOT Crude Oil

One way to determine whether the physical or chemical properties of crude oil influence pipeline accidents is to look at accidents that involved crude oil versus accidents that involved transporting other substances (e.g., natural gas). If there is a discernable difference between accidents for different substance types, we may be able to infer that there are inherent properties of crude oil (that don't exist / are less prominent in other substances) that are contributing to these incidents.

Approach #2 – Inherent Properties of Crude Oil

The second, and perhaps more direct approach to the problem, is to evaluate crude-oil specific incidents to see if its variations in composition (heavy vs. light, sour vs. sweet) are a direct contributor to pipeline accidents.

Considerations for Dataset Selection

Given that we have both Canadian and American datasets, it would be ideal to analyze at least one of each because:

- 1) The data that is required by the governing bodies may vary and therefore result in different attributes / information provided about the incident
- 2) Canada and the USA have varying geographies, and this may influence substance properties (e.g., difference in temperature could affect corrosion rates)
- 3) Because Canada and the USA have different governing bodies for their pipelines, the pipeline-specific specifications (e.g., what materials meet code to build pipelines) may vary, which could contribute to pipeline incidents based on the substance being transported

Additionally, it seems logical to select data that is as recent as possible and to select datasets with the highest number of records (assuming they are of decent quality).

DATASET COLLECTION

Given the desired approaches and considerations for dataset selection, dataset A (Canada) will be selected for **Approach #1** and dataset E (USA) for **Approach #2**.

Dataset E had to be used for **Approach #2** given that it is the only dataset that contains property-specific information about the crude oil being transported*. That left the two Canadian datasets for **Approach #1**; dataset A was selected because dataset B contained information about incidents at pipeline facilities / processing plants which is outside of the scope of this analysis.

*Dataset B contains crude oil vs. crude oil-sour, however, only two of the hundreds of crude oil records are crude oil-sour.

Analysis – Approach #1

The key steps taken in the analysis of **Approach #1** are summarized below. The corresponding script for this analysis is in the “dataset_A.ipynb” file, which falls under the “src” folder. The script itself includes further explanation of the decision-making process.

Step	Description
1. Data Ingestion	1.1 Load in dataset A given its relative location to the root directory.
2. Data Cleaning	2.1 Clean the dataframe so that it can be used in analysis. Use information from the dataset’s corresponding data dictionary 2.1 Identify columns from data dictionary that could be relevant to 1) the inherent properties of the substance being carried, 2) pipeline specifications, or 3) the cause of the incident. 2.2 Optimize the substance carried column using the released substance type column. Want to perform this task before dropping attributes that don’t meet a specific threshold. Additionally, map any substance that does not equal crude oil to 0, and any crude oil substance to 1. 2.3 Drop any attributes (columns) that are less than 50% populated. 2.4 Clean numeric columns 2.5 Map categorical columns to binary/numeric. 2.6 Map descriptive columns (e.g., “what happened category”) to binary columns, i.e., one-hot encode.
3. Data Exploration	3.1 Generate a profiling report of the dataframe to generate a heatmap, identify highly correlated pairs, or display errors / inconsistencies in the dataset (e.g., duplicate rows). 3.2 Output descriptive statistics (count, mean, etc.) of numeric columns to identify additional trends.
4. ML Models	4.1 Preprocess dataframe for ml models by running a correlation matrix and dropping one of the columns found in each high-correlated pair. 4.2 Run a logistic regression model on the data to identify if substance type (crude oil vs. not crude oil) can be predicted by the other attributes. 4.3 Random forest. 4.4 Decision tree. 4.5 Support Vector Machine (SVM).

Analysis – Approach #2

Similar to the previous slide, the key steps taken in the analysis of **Approach #2** are summarized below. The corresponding script for this analysis is in the “dataset_E.ipynb” file. Once again, the script itself includes further explanation of the decision-making process.

Step	Description
1. Data Ingestion	1.1 Load in dataset E given its relative location to the root directory.
2. Data Cleaning	2.1 Prioritize the commodity column as their properties (sweet vs. sour, heavy vs light) are the basis of this analysis.
	2.2 Map these properties to binary columns using string extrapolation. E.g., “Heavy Louisiana Sweet” would mark a 1 in the “heavy” and “sweet” columns. These properties were only given for a tiny portion (~45/1355) of the records.
	2.3 Identify and fill in the general cause of the incident using logic from the corresponding data dictionary.
	2.4 Clean the remaining columns using a similar process to that for dataset A / Approach #1.
3. Data Exploration	3.1 Generate a profiling report of the dataframe to generate a heatmap, identify highly correlated pairs, or display errors / inconsistencies in the dataset (e.g., duplicate rows).
4. ML Models	4.1 Preprocess dataframe for ml models by running a correlation matrix and dropping one of the columns found in each high-correlated pair.
	4.2 Run a decision tree model on the data to identify if the specific properties of the crude oil had an affect on the incident type.
	4.3 Random forest.
	4.4 Logistic regression.

Incomplete

The majority of the steps from 2.3 - onwards were performed and iterated upon in an attempt to salvage the approach. Ultimately, the minimal number of records (~45) with property-specific information on crude oil limited the extent of this analysis.

Results – Approach #1

Four machine learning models were run on the cleaned dataset from Approach #1. An interpretation of their results is given below.

Random Forest / Decision Tree

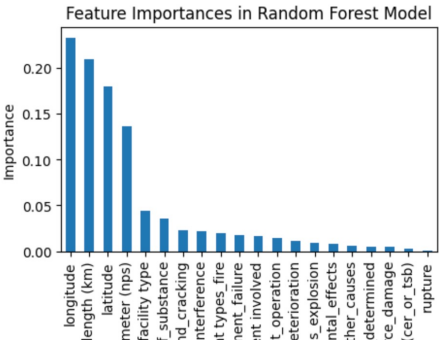
Logistic Regression / SVM

Actual	Predicted		ACCURACY: 0.95	
	NOT Crude Oil	Crude Oil	Precision	Recall
	NOT Crude Oil	82	0.95	0.99
Crude Oil	4	19	0.95	0.83

Feature Importance

After evaluating feature importance for the random forest, as seen below, it revealed that longitude and latitude, and pipeline length and diameter are the four biggest predictors of crude oil vs. not crude oil incidents.

This implies that crude oil pipelines exist at different locations than non-crude oil pipelines and that they have different pipeline dimensions.



DROPPING COLS & RE-RUNNING MODEL

After dropping these four columns (as they cannot be directly related to a substance-induced incident), the model produces mediocre results, similar to those of the logistic regression and SVM models.

Actual	Predicted		ACCURACY: 0.80	
	NOT Crude Oil	Crude Oil	Precision	Recall
	NOT Crude Oil	77	0.82	0.93
Crude Oil	17	6	0.50	0.26

Interpretation

Results: The models are good at predicting NOT Crude Oil but they are not good at predicting Crude Oil. This could be a result of the imbalanced dataset, given that there are ~2x as many NOT Crude Oil records as there are Crude Oil.

Coefficients: Similar coefficients are predicting Crude Oil vs. NOT Crude Oil. Given our domain knowledge, they do not seem to be especially relevant.

CONCLUSION

Given the model’s ability to accurately predict NOT Crude Oil incidents, we could conclude that there are inherent differences in NOT Crude Oil accidents vs. Crude Oil accidents. However, further analysis needs to be performed to identify exactly what properties of Crude Oil (or of NOT Crude Oil) are contributing to these accidents.

Results – Approach #2

Less distinct results from Approach #2 necessitate a discussion and ideas for future investigation.

Discussion

Dataset E was the only dataset found for Canada / USA that contained property-specific information for crude oil pipeline accidents. Upon extracting this property-specific information, fewer than 50 records remained. Attempts to enhance the data were made, such as keeping in crude oil incidents that did not have specific chemical or physical properties. But proceeding with the analysis of this small sample size ultimately led to unusable results. The data exploration profiling report revealed that light and sour attributes were highly correlated and that the sweet attribute occurred most frequently. However, given the limited sample size, we cannot confidently make conclusions from these findings.

Enhanced Data

- With a larger dataset, we could have executed a process like that of Approach #1 to determine what properties of crude oil (if any) influence pipeline accidents. This would consist of further cleaning of relevant attributes and iterating upon models.
- A beneficial dataset – which could not be found for this analysis – would contain information about the **successful transfer** of crude oil across pipelines. For example, if sour crude oil is being transferred the most frequently, but it is found in the least number of incidents, you could potentially conclude that sulfur is less responsible for crude oil-based pipeline accidents.
- A method to manually enhance the data is to research the properties of the specific crude oils found in the dataset; for example, researching “ALASKAN NORTH SLOPE CRUDE OIL” to identify if it is sour or sweet, heavy or light. This method was not used in this analysis as the pay-off was not justified: it would have taken significant time and would have only added ~50 records.

Isolating the Problem

There are myriad factors involved in pipeline incidents which makes it difficult to identify how the properties of crude oil are contributing to the incidents. To isolate the problem, a controlled study could be conducted. This would consist of using various pipeline materials and sizes, running specific crude oils through them (crude oil – sour, crude oil – sweet, crude oil – sour and heavy, etc.), collecting the data, and performing a regression analysis to determine which components of the crude oil are affecting corrosion and wear and tear on the pipelines.

Conclusion & Suggested Next Steps

Given the results from our two approaches, we cannot safely conclude which properties of crude oil, if any, are responsible for pipeline accidents. But we can provide four distinct next steps, in order of importance, that could facilitate further analysis of this topic.

Conclusion

From Approach #1, we concluded that there are potential differences between pipeline accidents in which crude oil was the substance being transferred vs. other substances (e.g., natural gas). From Approach #2, we were unable to form a confident conclusion given our limited dataset. However, the collective analyses from these two approaches provided insight into how this problem can be isolated and investigated to a further extent.

1 – CONTROLLED STUDY

Given the complexity of pipeline incidents, the number one suggestion for future investigation is isolating the problem in a controlled environment. A controlled study would take significant time and resources but would provide the most accurate results as to how the physical and chemical properties of crude oil influence pipeline accidents. (Similar study: [9])

3 – ALL PIPELINE TRANSFER DATA

Instead of only looking at pipeline incidents, it could prove valuable to look at all data surrounding the transfer of substances via pipelines. This could identify the proportion of crude oil properties that are most-involved in pipeline accidents. Once again, contacting the governing bodies of pipeline regulations would be a good first step for sourcing this data.

2 – LARGER DATASET FOR CRUDE OIL INCIDENTS

Other than performing a controlled study, the most direct approach for determining which properties of crude oil affect pipeline incidents is to identify how different types of crude oil relate to the frequency and type of pipeline accidents. Two methods to gather this data are: 1) reaching out to the governing bodies (CER / PHMSA) for additional data or 2) manually researching (or contacting) each company's crude oil to determine its properties.

4 – EVALUATE FEATURE IMPORTANCE OF “NOT CRUDE OIL” INCIDENTS

Given that the models from Approach #1 were able to confidently predict NOT Crude Oil incidents, it would be worthwhile to evaluate which features are contributing to these predictions. This could reveal which attributes are relevant for NOT Crude Oil instances, which could be potentially helpful for their Crude Oil counterparts.

Additionally, the model could be expanded by concatenating / merging additional datasets. This is not recommended as a first approach due to inconsistencies in data structures.

Part B – Regulations & Implications

Current & Proposed Regulations

To propose regulations, we need to understand the current regulations and how they’ve evolved over time.

Questions:	1. What regulations on those properties would you recommend? 2. Have we gotten better at regulating those properties over time?
Governing Body, Year	Regulation
PHMSA, 2013	The PHMSA along with Federal Railroad Administration launched Operation Classification to verify that crude oil is being properly classified by performing unannounced spot inspections and data collection, and by sampling and testing at strategic locations in the crude oil transportation chain [10].
DOT / Transport Canada, 2015	The US Department of Transportation (DOT) in conjunction with Transport Canada created a plan to conduct a research study to develop a more comprehensive understanding of the properties of crude oil [10]. The review has a heavy focus on identifying which properties are correlated with combustion. The study discusses a difficulty in utilizing available data because of sample point variability, sampling method variability, and analytical method variability. As of the plan’s announcement, there was no funding for the work (and the subsequent product cannot be found online).
AER, 2023	The Alberta Energy Regulator (AER) makes frequent updates to the OIL AND GAS CONSERVATION RULES (among other documents). With regards to the regulation of crude oil, the current version of this document states that the operator of wells / oil sands are required to keep records of the crude oil received into the pipeline and file the records with the Regulator [12].

DISCUSSION

The governing bodies of crude oil transportation across Canada and the USA have regulations in place to monitor the chemical and physical properties of crude oil. However, given the quantity of crude oil that is being transferred across North America, and the fact that crude oils have unique compositions, it seems unreasonable that a governing body could keep track of the properties of every crude oil that they govern. This is why an audit-based approach, like the PHMSA’s random sampling and AER’s required documentation, form a reasonable set of regulations.

Regarding research studies concerning the composition of crude oil, it seems as though numerous organizations have proposed scientific plans to identify relevant properties ([1], [10]). But the necessary funding to execute these studies is not being given. This could potentially be a result of substance-related incidents making up a small portion of total incidents (e.g., human error contributes to more incidents than internal corrosion-related incidents).

Ultimately, the regulations regarding the properties of crude oil do not seem to have improved much over time. The intention is there, but the execution is not. This conclusion is reinforced by the datasets provided by these governing bodies on pipeline incidents – next to no data is given regarding the composition of the crude oil that is involved in the respective accident.

Without knowing which properties of crude oil are influencing pipeline incidents, we cannot make property-specific recommendations. However, we can make recommendations regarding the sampling and disclosure of crude oil across pipelines (for both regular operations and within incidents). It seems reasonable to require that when a pipeline incident occurs, the company samples the crude oil at the point of failure and includes this in their disclosure. Additionally, it does not seem unreasonable to require sampling of substances during regular operations at frequent intervals (instead of just at the start) along pipelines that carry crude oil. These could be disclosed at quarterly or yearly intervals.

Quantifying Success & Market Implications

Should the proposed regulations be implemented, we need to understand how they could impact the oil market and there needs to be a method to quantify their success.

Questions:

1. What implications on the oil market would these regulations have?
2. How can success of these regulations be quantified in the short term and long term?

Metrics for Quantifying Success

SHORT TERM

- Meet funding requirements for further scientific evaluation on the inherent properties of crude oil that are influencing pipeline accidents.
- Have companies (or governing bodies) take samples of and disclose the crude oil involved in x percentage of pipeline incidents where x increases each year and eventually converges to 1.
- Have companies (or governing bodies) take samples of and disclose the crude oil within their pipelines every x kms and at higher-risk points for their pipelines that transfer crude oil. Determine x based on the pipeline length.

LONG TERM

- Identify which properties of crude oil are leading to pipeline accidents; extract or minimize the transfer of these properties.
- Minimize the number of crude oil-related pipeline incidents based on the analysis of the additional sampling and disclosure of data. Predict the ideal response type for an incident based on the substance that was being carried.
- Analyze trends in crude oil prices to identify any economic impact of these regulations (however, safety is the number one priority).

Implications on the Oil Market

SHORT TERM

- Additional funding required to meet disclosures
 - This would require an up-front investment and could influence company profitability
 - This could also influence the price of oil that trickles down to consumers
- Additional sampling could slow down supply chain which could increase global oil prices (e.g., oil supply chain during COVID-19)

LONG TERM

- Improved safety of pipelines could lead to better public perception of oil transportation. This is especially crucial with a portion of the public demanding the switch to clean energy.
- Enhanced transparency regarding crude oil properties could reduce competitive advantages in the market, leading to a more balanced playing field.

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Format: Author's Name, Title, URL

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An aerial photograph showing a vibrant blue river with white rapids flowing through a dense green forest. A paved road runs along the right bank of the river. The word "Appendix" is written in white serif font on the left side of the image, with a horizontal line extending from its right edge.

Appendix

Appendix A: Composition of Bakken Crude Oil

Hydrocarbon gas	Concentration in Liquid Volume %	DOT Classification
Methane	<0.01	Div. 2.1 Gas
Ethane	0.05	Div. 2.1 Gas
Propane	0.80	Div. 2.1 Gas
Iso-Butane	0.46	Div. 2.1 Gas
N- Butane	2.36	Div. 2.1 Gas
Total Gas	3.67	
Neopentane	<0.01	Class3; PG I
Iso-Pentane	1.33	Class3; PG I
N-Pentane	2.36	Class3; PG II
Hexanes	4.10	Class3; PG II
Heptanes Plus	88.56	Varies Class 3; PG II to combustible liquid to non-regulated
Benzenes	0.08	Class 3; PG II